How to write a thesis that benefits you

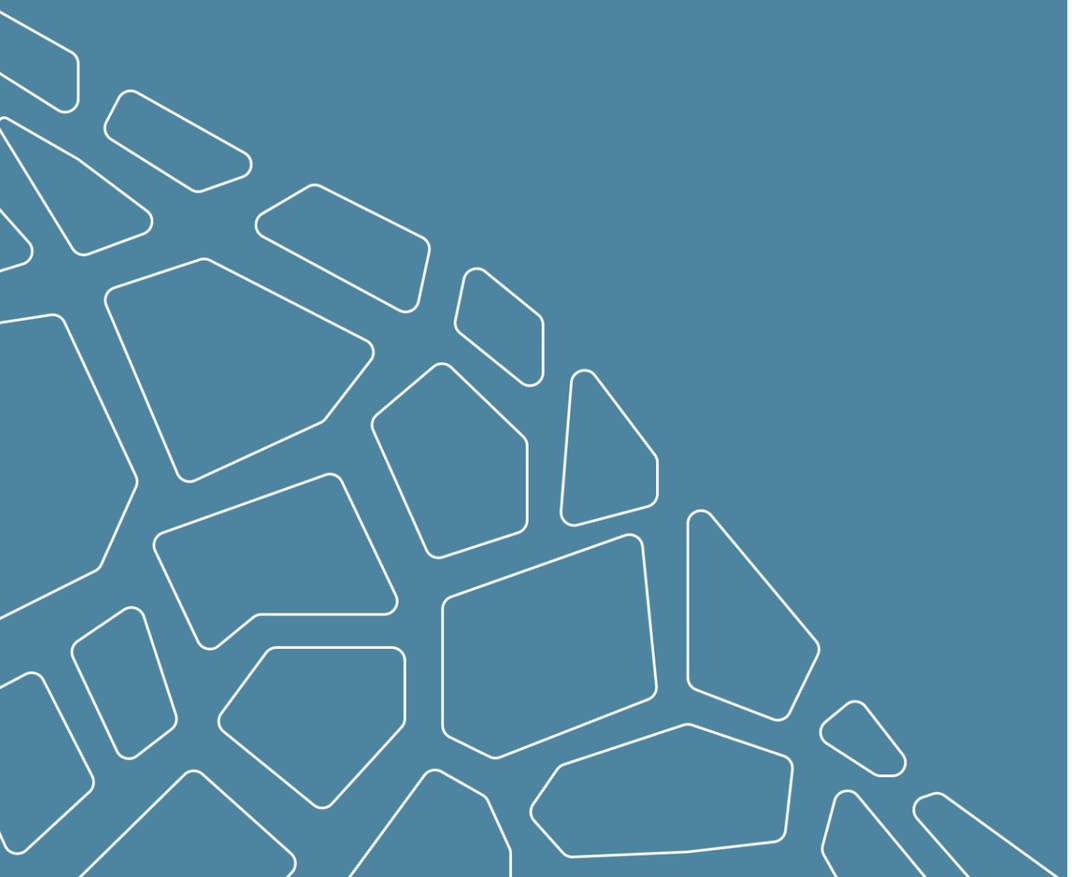
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Senior researcher

girafe ai

MIPT, fall 2026

Outline



1. Motivation
2. Resources
3. Steps of work

Disclaimer

- Based on my personal experience
  - doesn’t limit your ways to do things
- Applicable to Machine Learning
  - and most STEM disciplines
- For some of you many points may sound obvious
  - but try to apply them and you will win

Thesis is different from what you had before

It is not a pre-build road which you drive just pushing a gas pedal on a given car (which courses were like).

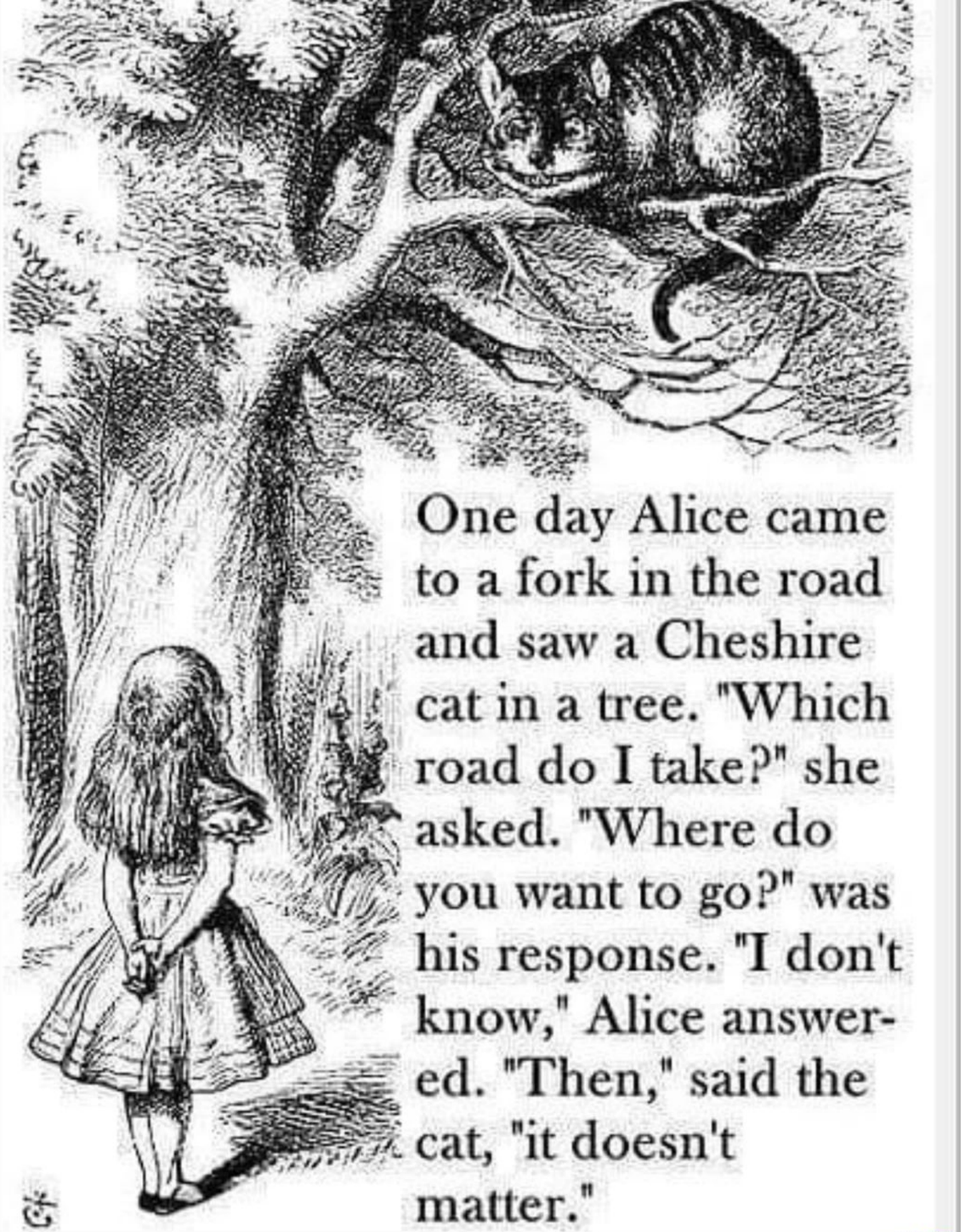
It’s more like walking in woods by yourself.

You have to take tools with you and be able to construct everything you need from improvised means around you.





Advice from a mathematician of the past



Motivation

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01

What are the functions thesis?

- Dive deep into science subfield
- Show your performance and skills on a new unknow subfield
- Get practical (maybe industrial) experience, line in a resume
- Contributing to science (or any knowledge)
- Present your results to professionals
- Publish a scientific article
- Get self-confidence in ML
- Prepare to PhD (continue to develop current topic)

Core function:

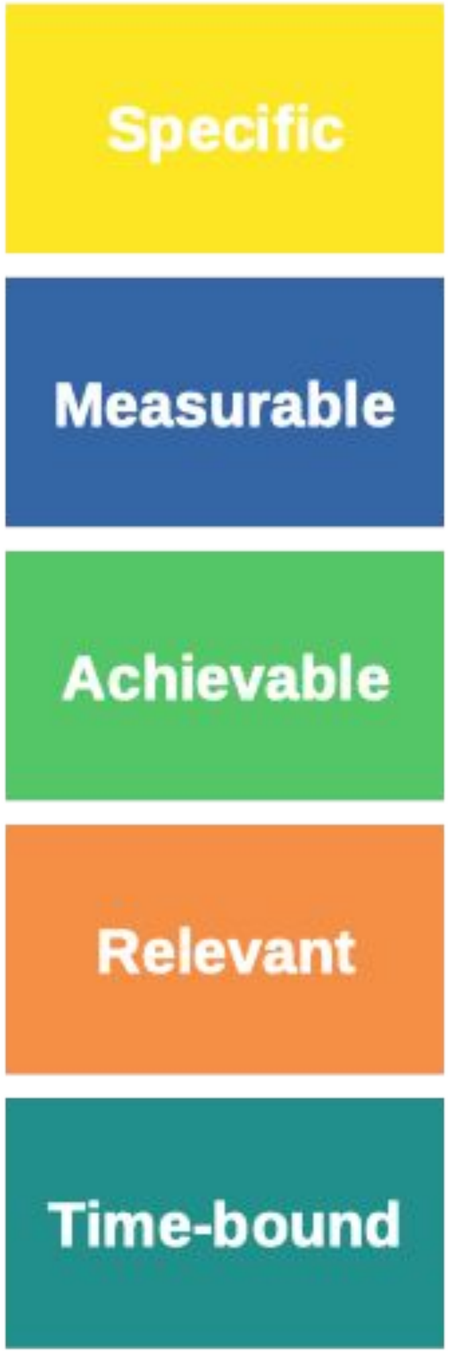
Qualification work to receive a degree

Goal setting

SMART is one widely used frameworks to set up a goal.

It is applicable to thesis as well.

Also there are other frameworks for this such as Objectives and key results (OKR)



Typical goals

- Receive diploma
- Submit scientific result
  - talk or poster on conference
  - article in journal
- Practical experience
  - release to production
  - open source project
  - get a job

You need to **write it down explicitly** before start of the work.

Although your goal can change during term, it is not recommended

Learning about learning

https://www.youtube.com/watch?v=0xS68sl2D70

Slow and fast thinking

Why use textbooks instead of video materials?
https://www.youtube.com/watch?v=oEQaENxv70M

How to use textbooks?
https://www.youtube.com/watch?v=_CKa2_wxeYQ

Resources

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02

Resources to consider

- Your time
- Source and amount of data
- Compute

Time

Define how much time will you spend on thesis work each week.

Most efficient way is to work uniformly over all period.

Gradation is the following (hours a week):

- 0-2 - usually poor results
- 2-4 - limited potential, modest goals and result
- 5-8 - some potential, average result
- >8 - promising amount of work, chance for a good thesis

When to start?

As soon as possible

You will go through several stages and first part requires only searching for information and reading papers, so you can start today.

Usually best performing students spend a year on thesis.
Good results also can be done it half a year.
Over half a year of work is usually results to poor performance.

Deadlines

- Topic and literature overview
  - January
- Main experiments finish
  - April
- Text and presentation
  - May

Data and compute

Better to have your custom data source rather than open datasets (but don’t neglect them).
You can use proprietary data, no need to publish data.

Compute is also important resource and you’d better have dedicated one (your hardware, cloud, etc.).
For compute resources it’s good idea to buy computer for a group of students (or professionals) and use it together.

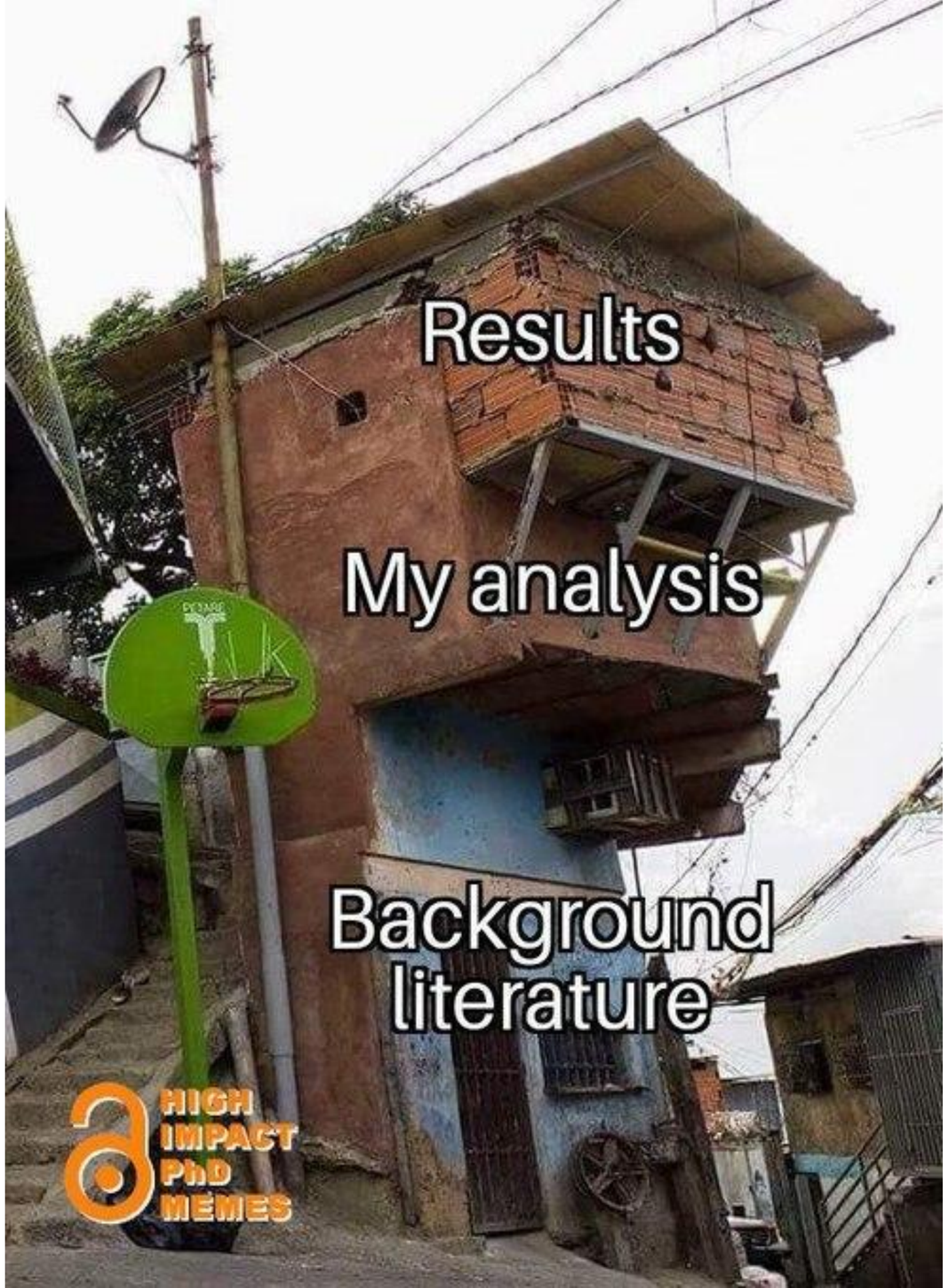
Steps of work

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03

Steps of work

- Define a motivation
- Find a topic
- Find a thesis advisor
- Create literature overview
- Find data
- Build a baseline
- Perform experiments
- Create presentation
  - also text
- Publish a paper/poster
- Receive a diploma! 💫



Examples of inspired thesis works

- colorifying comics
- music generation
- monocular deph estimation
- powerplant consumption prediction
- postcards image database with search
- skin cancer detector

https://github.com/girafe-ai/journal-club

An Opinionated Guide to ML Research

Research advice from OpenAI cofounder, PPO training method in RL author John Schulman to OpenAI Fellows program.

http://joschu.net/blog/opinionated-guide-ml-research.html



Understanding is the new bottleneck

Understanding is the key in the new **agent first world**!

https://www.geoffreylitt.com/2026/07/02/understanding-is-the-new-bottleneck.html

Thanks for attention!

Questions?



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